Smart Document Parser: Resume Entity Extraction

Project Report (Assignment 3)

GitHub Repo: https://github.com/Akshat-A-K/Resumer-Parser

1. Abstract

We built a resume parser that converts PDF or image resumes into clean JSON. The system uses OCR and PDF text extraction, and a local LLM (Ollama with Llama 3.2) to extract entities like name, email, skills, education, and experience. A Streamlit web app lets the user upload a file, review the output, edit it, and download the final JSON or CSV. We also created an evaluation mode that compares predictions with a labeled dataset and reports precision, recall, and F1 scores. This report explains the problem, dataset, method, and results for two approaches. The code is in the branches: ollama-basic and prompt-based-fine-tuning.

2. Introduction

In hiring, companies receive many resumes for one job opening. Manually reading each resume and writing the details is slow. So we need an automated system that can read a resume and extract important data in a structured format. The assignment asks for a smart document parser, and we selected the resume parsing task. Our system is fully local and works without paid APIs.

3. Problem Statement and Objectives

The system should take a resume in PDF or image form and return a structured JSON with key information. It should handle different layouts and noisy text and still give usable results.

We allow both PDF resumes and image resumes (PNG, JPG, JPEG, BMP, TIFF, WEBP). For images we use OCR, and for PDFs we extract text directly.

Main objectives:

- Extract contact, education, experience, skills, and projects.
- Provide a simple UI for upload, review, edit, and download.
- Provide evaluation results on labeled data.

4. Dataset

We used the Resume Entities for NER dataset from Kaggle [1]. It has 220 annotated resume samples in newline-delimited JSON format. Each sample has resume text and labeled entity spans. The labeled fields include Name, Email Address, Location, Designation, Companies worked at, Skills, College Name, Degree, Graduation Year, and Years of Experience. Many fields in our schema (such as phone, links, projects, or certifications) do not have ground truth in the dataset, so evaluation is only for the mapped 10 fields.

5. Approach 1: Streamlit Application (ollama-basic)

This approach is in the branch ollama-basic. The pipeline is:

1. User uploads a PDF or image.
2. Text is extracted using PDF parsers for PDF, and OCR for images.
3. A strict JSON schema prompt is sent to the local LLM.

4. The output is cleaned and validated into the schema.

5. Results are shown in the UI, user can edit, and then download JSON or CSV.

How the Streamlit app looks and how it is used for ML work: the interface is simple and clean, with a left sidebar to choose the mode (Parse One Resume or Evaluate Quality). In the main area, the user sees the file upload, the extracted text preview, and the editable output. For ML work, it is useful because it gives fast feedback, lets us correct output, and makes it easy to test many samples without extra scripts.

Key points of this approach:

- Uses local Ollama with Llama 3.2 model and temperature 0.
- Uses a single prompt with selected fields.
- Results are validated so empty or missing values are handled safely.
- Evaluation mode runs on a chosen number of labeled samples.

6. Models and Tools Used

- Streamlit for the UI [2].
- PyMuPDF with pypdf fallback for PDF text extraction [5].
- EasyOCR with Pillow and NumPy for image text extraction [7].
- Ollama for local LLM inference (Llama 3.2, model must be downloaded) [3, 4].
- Pydantic for schema validation and normalization [6].
- Pandas for evaluation tables.

7. Approach 2: Prompt-Based Fine-Tuning (prompt-based-fine-tuning)

This approach is in the branch prompt-based-fine-tuning. It focuses on stronger prompt engineering and structured extraction. We include it here for comparison.

Key ideas:

- Section-wise prompting: resume text is split into sections like education, experience, projects, and skills. Each section is parsed with only relevant fields.
- Extended schema: structured lists are used for education, experience, and projects.
- Extra normalization: JSON repair, retries, and link mapping to improve output quality.
- Caching: repeated extractions are faster.

This method aims to improve list-heavy fields by reducing prompt noise and focusing on relevant text for each section.

8. Evaluation Methodology

We used the labeled dataset to measure accuracy. For string fields, we compute exact match, fuzzy match, and token-based precision/recall/F1. For list fields, we use set-based precision/recall/F1 with fuzzy matching and Jaccard similarity. We also report macro-averaged precision, recall, and F1 across all fields.

9. Results for Streamlit Approach (10 Samples)

Latest evaluation for the ollama-basic branch was run on 10 samples. Macro scores:

- Precision: 0.5712
- Recall: 0.5883
- F1: 0.5561

Entity Type	Precision	Recall	F1	Support
Name	0.90	0.90	0.90	9
Graduation Year	0.40	0.40	0.40	7
Years of Experience	0.50	0.50	0.50	2
Email	0.70	0.60	0.6333	8
Location	0.45	0.80	0.5633	8
Degree	0.80	0.625	0.6857	8
Designation	0.70	0.65	0.6667	8
Companies Worked At	0.2417	0.50	0.3067	8
Skills	0.15	0.20	0.1667	10
College Name	0.87	0.7083	0.7389	10

10. Results for Prompt-Based Fine-Tuning Branch (10 Samples)

Reported macro scores from the prompt-based-fine-tuning branch:

- Precision: 0.585
- Recall: 0.617
- F1: 0.575

Entity Type	Precision	Recall	F1	Support
Name	0.900	0.900	0.900	9
Email	0.700	0.600	0.633	8
Location	0.500	0.900	0.630	8
Graduation Year	0.400	0.400	0.400	7
Years of Experience	0.600	0.550	0.566	2
College Name	0.783	0.725	0.745	10
Degree	0.800	0.650	0.700	8
Designation	0.600	0.583	0.550	8
Companies Worked At	0.541	0.766	0.586	8
Skills	0.028	0.100	0.044	10

11. Comparison and Discussion

Macro-average comparison (10 samples each):

Approach	Precision	Recall	F1
ollama-basic	0.5712	0.5883	0.5561
prompt-based-fine-tuning	0.5850	0.6170	0.5750

Observations in simple words:

- Both approaches are strong in name and education fields.
- The prompt-based branch gives slightly higher macro scores.
- Prompt-based method takes more time because it runs more steps and more prompts.
- Skills and companies are the hardest fields in both approaches.
- Section-wise prompting helps in some list fields, but skills are still weak because of different naming styles.

12. Limitations

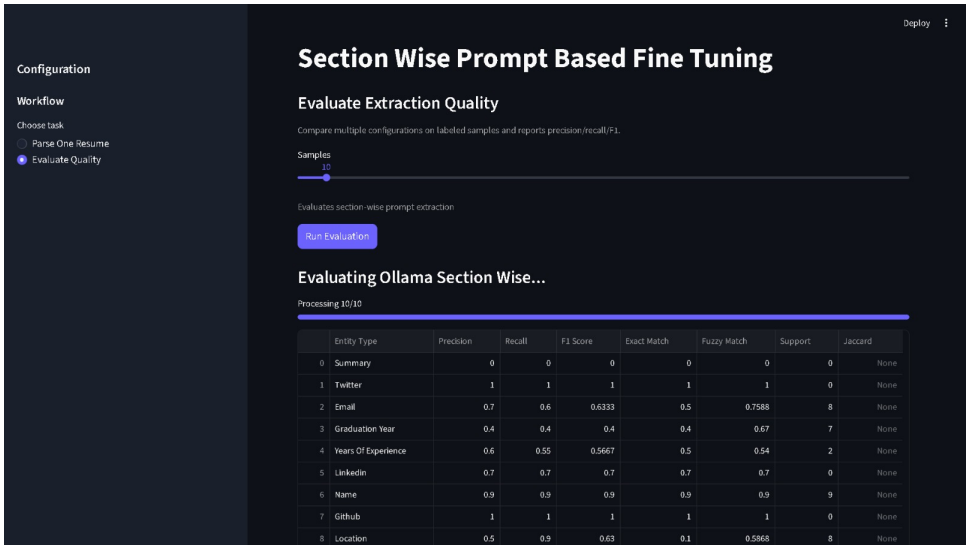
- The evaluation used only 10 samples, so results can change with more data.
- OCR quality depends on resume image quality.
- Some fields are not in the dataset, so they are not evaluated.

13. Conclusion and Future Work

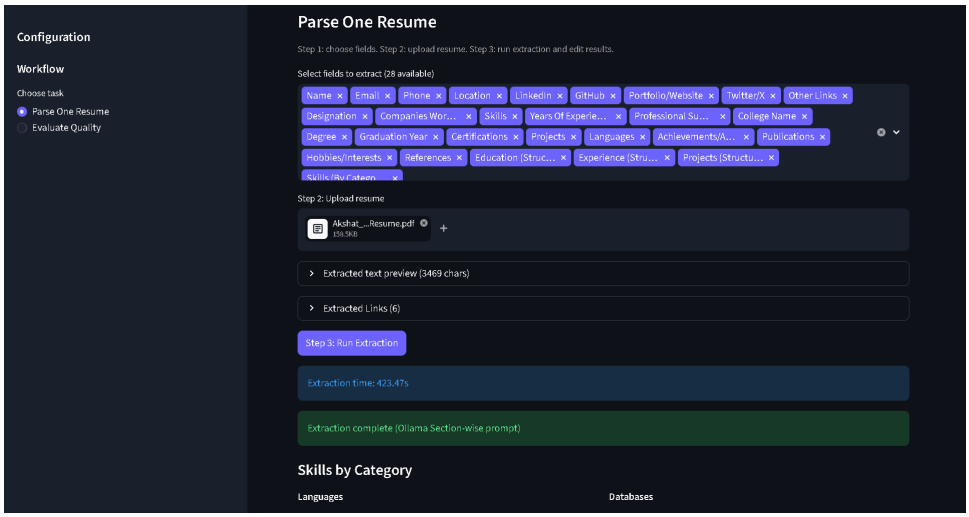
We built a working resume parser with a Streamlit app, and we also tested a prompt-based fine-tuning approach. The ollama-basic approach is simpler and stable. The prompt-based-fine-tuning approach gives slightly better macro scores on the same sample size, but takes more time. For future work, we should run larger evaluations, improve skill normalization, and try a larger model for better accuracy.

14. Screenshots

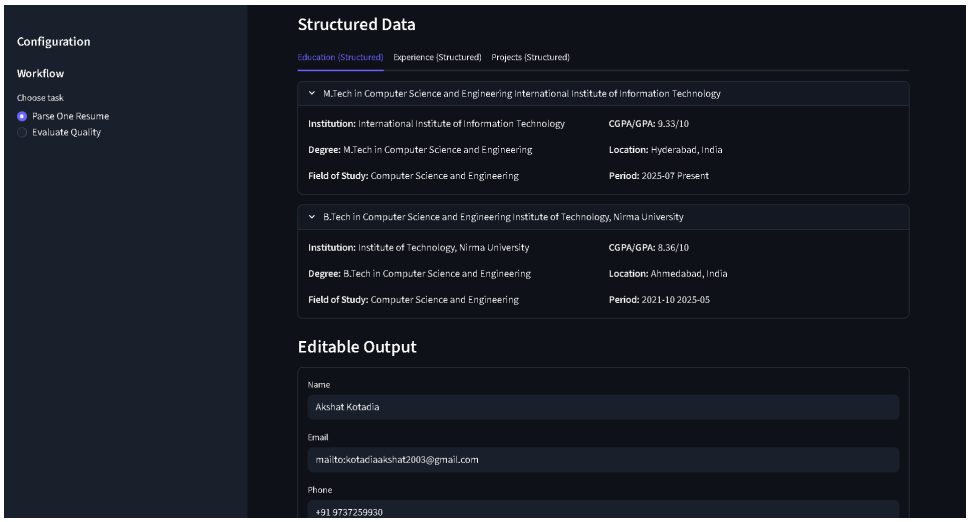
Below we include the evaluation view and two stacked views from Parse One Resume.



Evaluate Quality mode (section-wise prompt evaluation)



Parse One Resume: field selection and upload



Parse One Resume: structured output and editable fields

References

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[6] Samuel Colvin et al. Pydantic Documentation.
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[7] JaidedAI. EasyOCR: Ready-to-Use OCR with Deep Learning. GitHub Repository.
https://github.com/JaidedAI/EasyOCR